Project 3 Status Update and Retrospective – Gang_X1

From: Hrudith Lakshminarasimman
To: Dr. Paul Taele and TA Dineth Gunawardena
Subject: Project 3 Individual Status Report and Retrospective

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1. Introduction

I am writing to provide my individual status report and retrospective for **Project 3** as a member of **team Gang_X1**.
This report summarizes the **project purpose**, **my role and contributions**, **shared work**, **current progress**, **challenges**, and **reflections**.

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2. Project Summary

Our project delivers a complete **Sharetea SaaS frontend system** that integrates the following interfaces:

- Manager
- Cashier
- Customer Kiosk
- Kitchen Display
- Menu Board

It extends the **Project 2 POS backend** and introduces new features including:

- Accessibility support
- Multilingual translation

- Weather-aware functionality
- OAuth authentication

All features align with the **course specifications** and **Project 3 requirements**.

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3. Project Objectives and Accessibility Requirements

The system is designed to:

- Provide a **modern, accessible, multi-role interface stack**.
- Improve **operational clarity** for all user roles.
- Support **bilingual users**.
- Deliver a **kiosk interface compliant with WCAG 2.1** standards.

Accessibility testing was performed according to the assigned personas: **Maria**, **Vishnu**, and **Carol**.

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4. Team Roles and Responsibilities

Team Member	Role	Key Responsibilities
Hrudith (Myself)	Full Stack Developer	Architecture, frontend-backend integration, implementation, debugging, and code reviews
Jhanghir	Backend Developer	Database schema, authentication, PostgreSQL design
Sudeep	Backend Developer	API endpoints, orders, and inventory logic
Rahul S.	Frontend Developer	React components, UI implementation

Rahul G.	Database & API Engineer	SQL optimization, migration scripts
Nicholas	UI/UX & Testing	Accessibility testing, persona validation, wireframes

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5. My Individual Contributions

- Led **full-stack integration** between Vite React frontend and Node/Express backend.
- Implemented **role-based routing**, session handling, and multi-interface structure.
- Integrated external APIs: **OAuth**, **Google Translate**, and **OpenWeather**.
- Built **cashier ordering flows**, **menu board data pipelines**, and **shared UI components**.
- Conducted **debugging** and **cross-team code reviews**.
- Supported **accessibility implementation** (ARIA roles, contrast adjustments, scalable UI structures).
- Contributed to **Sprint 1 and Sprint 2 planning** and **backlog refinement**.

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6. Shared Work and Contribution Percentages

Shared Task	Team Members	My Contribution	Description
OAuth + Auth Routing	Me, Jhanghir	55%	Implemented OAuth flow, route guards, and session persistence
Menu + Inventory Data Pipeline	Me, Sudeep, Rahul G.	40%	Connected PostgreSQL schema to frontend fetch layer
Cashier UI + Order Flow	Me, Rahul S.	50%	Implemented layout, actions, and state integration
Accessibility Integration	Me, Nicholas	35%	Ensured font scaling, contrast modes, and ARIA compliance

Kiosk Navigation + Structure	Me, Nicholas, Rahul S.	30%	Defined frame structure, routing, and layout skeleton

7. Contribution Ratings

All ratings average to **“equal”** across the team as per assignment instructions.

Team Member	Contribution Rating
Hrudith (Me)	Equal
Jhanghir	Equal
Sudeep	Equal
Rahul S.	Equal
Rahul G.	Equal
Nicholas	Equal

8. Status Update

Items I Completed

- Full-stack integration of **React + Node architecture**
- **Role-based routing** for Manager, Cashier, Kiosk, Kitchen Display, and Menu Board views
- **OAuth 2.0 authentication system**
- **Translation integration framework** and UI placement
- **OpenWeather API** consumption and contextual weather-based features
- **Cashier POS interaction logic** and order construction
- **Shared UI component library**

- **Accessibility scaffolding**, including scalable text, contrast presets, and ARIA mapping

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9. Challenges and Roadblocks

- **Accessibility Alignment:**
  The MVP user study revealed issues with visual affordances, low contrast, and unclear navigation hierarchy. These were addressed in Sprint 2.

- **API Rate Limits and Caching:**
  Required caching and local fallback strategies for Google Translate and weather data.

- **State Management Across Multiple Views:**
  Synchronizing kiosk, cashier, and menu board states required redesign of shared state management.

- **Deployment Stability:**
  Early Vercel deployments needed multiple environment adjustments.

All challenges were **resolved** or have **clear resolution plans for Sprint 3**.

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10. Personal Retrospective

I experienced significant growth in **full-stack architectural design**, particularly in integrating frontend state systems with backend APIs for a multi-role application.
The depth of technical work across different interfaces strengthened my **design**, **debugging**, and **system-thinking skills**.

Working in a **six-person team** presented synchronization challenges and stylistic differences, but our **backlog-driven workflow** ensured effective collaboration.

In future projects, I aim to:

- Implement **structured documentation** early.

- Define **interface contracts** at the start of development.

- Reduce **integration friction** to improve **velocity** and minimize **refactoring**.

11. Peer Evaluation Retrospective

The **anonymized peer evaluations** promoted accountability and improved communication throughout sprints.
They encouraged clarity in task ownership and proactive issue resolution.

However, differences in **task visibility**—especially between backend and frontend—made evaluations occasionally challenging.
Future iterations could benefit from:

- **Mid-sprint checkpoints**
- **More granular task attribution**

Overall, the process improved **team dynamics**, **collaboration quality**, and **balanced contributions**.

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12. Conclusion

This project significantly enhanced my technical, collaborative, and architectural skills.
It also reinforced the importance of accessibility, structured workflows, and cross-role communication within a full-stack development environment.

Sincerely,
Hrudith Lakshminarasimman